

Supplementary proofs for SPARQL_{circ}: Probabilistic Query Answering with Provenance Circuits

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This document contains the full proofs of the results stated in SPARQL_{circ}: *Probabilistic Query Answering with Provenance Circuits*. Definition, lemma, theorem and equation numbers refer to the numbering of that paper.

A PROOFS

PROOF OF LEMMA 4.5. Let g and g' be congruent. We prove $\llbracket g \rrbracket = \llbracket g' \rrbracket$ by cases on their common kind. For leaves, congruence gives $\lambda(g) = \lambda(g') = x$, and Definition 4.3 gives

$$\llbracket g \rrbracket = x = \llbracket g' \rrbracket.$$

For $\circ \in \{\oplus, \otimes\}$, let $\star_\circ$ denote $\vee$ for $\circ = \oplus$ and $\wedge$ for $\circ = \otimes$. Congruence gives $ch(g) = ch(g')$, hence

$$\llbracket g \rrbracket = \star_\circ \llbracket c \rrbracket = \star_\circ \llbracket c \rrbracket = \llbracket g' \rrbracket.$$

For difference gates, congruence gives $pos(g) = pos(g')$ and $neg(g) = neg(g')$. Therefore,

$$\begin{aligned} \llbracket g \rrbracket &= \left(\bigvee_{c \in pos(g)} \llbracket c \rrbracket \right) \wedge \neg \llbracket neg(g) \rrbracket \\ &= \left(\bigvee_{c \in pos(g')} \llbracket c \rrbracket \right) \wedge \neg \llbracket neg(g') \rrbracket \\ &= \llbracket g' \rrbracket. \end{aligned}$$

Replacing an incoming reference to g' by g preserves the parent formula. Induction over a topological order from the replaced gate to every answer root preserves all root formulas. $\square$

PROOF OF PROPOSITION 4.9. Let V_Q be the variables of the basic pattern, let $Z = V_Q \setminus W$, and let $\pi = (z_1, \dots, z_k)$ be the elimination order. For each triple pattern t_j , let

$$f_j(\rho) = \begin{cases} x_{\rho(t_j)}, & \rho(t_j) \in G, \\ \perp, & \rho(t_j) \notin G. \end{cases}$$

For a total binding $\theta : V_Q \rightarrow \text{adom}(G)$, let

$$F(\theta) = \bigwedge_{j=1}^m f_j(\theta|_{\text{scope}(f_j)}).$$

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For an output mapping μ , the flat plan denotes

$$\Phi_0(\mu) = \bigvee_{\substack{\theta: V_Q \rightarrow \text{adom}(G) \\ \theta|_W = \mu}} F(\theta). \quad (12)$$

Let $\mathcal{F}_i$ be the factors after eliminating $z_1, \dots, z_i$. For a binding ρ of the remaining variables, let

$$I_i(\rho) = \bigvee_{\sigma: \{z_1, \dots, z_i\} \rightarrow \text{adom}(G)} \bigwedge_{f \in \mathcal{F}_0} f((\rho \cup \sigma)|_{\text{scope}(f)}). \quad (13)$$

We prove that the conjunction of the factors in $\mathcal{F}_i$ at ρ equals $I_i(\rho)$. For $i = 0$, Equation (13) contains the unique empty assignment and equals the original factor conjunction. Assume the claim for $i - 1$. Let $\mathcal{A}_i \subseteq \mathcal{F}_{i-1}$ contain exactly the factors whose scope contains z_i , and let $U_i = (\bigcup_{f \in \mathcal{A}_i} \text{scope}(f)) \setminus \{z_i\}$. For $a \in \text{adom}(G)$, let $\rho_a = \rho \cup \{z_i \mapsto a\}$. Distributivity gives

$$\begin{aligned} I_i(\rho) &= \bigvee_{a \in \text{adom}(G)} \left[\bigwedge_{f \in \mathcal{A}_i} f(\rho_a|_{\text{scope}(f)}) \wedge \bigwedge_{f \notin \mathcal{A}_i} f(\rho|_{\text{scope}(f)}) \right] \\ &= h_i(\rho|_{U_i}) \wedge \bigwedge_{f \notin \mathcal{A}_i} f(\rho|_{\text{scope}(f)}), \end{aligned}$$

which is the conjunction after replacing $\mathcal{A}_i$ by h_i . At $i = k$, restriction to μ yields Equation (12), so both plans produce equivalent roots.

The semi-join of the factored plan drops a row of a base factor only when some remaining pattern t_i containing a constant has no row agreeing with it on their shared variables. Every total binding θ extending a dropped row then has $\theta(t_i) \notin G$, so $f_i(\theta) = \perp$ and $F(\theta) = \perp$. The dropped rows contribute only $\perp$ to Equation (12), which the semi-join therefore leaves unchanged.

The induced-width bound gives $|U_i| \leq w$. Each replacement factor has at most $|\text{adom}(G)|^w$ rows, while its product gates range over at most $|\text{adom}(G)|^{w+1}$ assignments to $U_i \cup \{z_i\}$. Since $|\text{adom}(G)| \leq 3|G|$ and there are $O(m)$ base factors and elimination steps, the gate count is $O(m(3|G|)^{w+1})$, which is $O(m|G|^{w+1})$ for fixed w . $\square$

PROOF OF LEMMA 4.10. We use structural induction on e ; the induction claim ranges over all $u, v \in D_G$ and all $W \subseteq G$.

IRI. Let $e = p$. Definition 4.3 gives

$$v_W(\llbracket b_{uv}^p \rrbracket) = \top \iff \bigvee_{t=(u,p,v) \in G} v_W(x_t) = \top$$

$$\begin{aligned} &\iff (u, p, v) \in W \\ &\iff (u, v) \in \text{Rel}_p(W). \end{aligned}$$

The empty child set yields $\perp$ on both sides.

Inverse. Let $e = \widehat{e}_1$. By construction and the induction hypothesis,

$$\begin{aligned} v_W(\llbracket b_{uv}^{e_1} \rrbracket) = \top &\iff v_W(\llbracket b_{vu}^{e_1} \rrbracket) = \top \\ &\iff (v, u) \in \text{Rel}_{e_1}(W) \\ &\iff (u, v) \in \text{Rel}_{\widehat{e}_1}(W). \end{aligned}$$

Alternative. Let $e = e_1 \mid e_2$. Since $\llbracket g_\perp \rrbracket = \perp$, omitting $g_\perp$ does not change the disjunction. The two induction hypotheses yield

$$\begin{aligned} v_W(\llbracket b_{uv}^{e_1 \mid e_2} \rrbracket) = \top &\iff v_W(\llbracket b_{uv}^{e_1} \rrbracket) = \top \vee v_W(\llbracket b_{uv}^{e_2} \rrbracket) = \top \\ &\iff (u, v) \in \text{Rel}_{e_1}(W) \vee (u, v) \in \text{Rel}_{e_2}(W) \\ &\iff (u, v) \in \text{Rel}_{e_1 \mid e_2}(W). \end{aligned}$$

Sequence. Let $e = e_1/e_2$. Equation (4), Definition 4.3, and the induction hypotheses give

$$\begin{aligned} v_W(\llbracket b_{uv}^{e_1/e_2} \rrbracket) = \top &\iff \exists w \in D_G : v_W(\llbracket b_{uw}^{e_1} \rrbracket) = \top \\ &\quad \wedge v_W(\llbracket b_{vw}^{e_2} \rrbracket) = \top \\ &\iff \exists w \in D_G : (u, w) \in \text{Rel}_{e_1}(W) \\ &\quad \wedge (w, v) \in \text{Rel}_{e_2}(W) \\ &\iff (u, v) \in \text{Rel}_{e_1/e_2}(W). \end{aligned}$$

If no midpoint satisfies the first line, the outer disjunction is empty and both sides are false. These cases exhaust the grammar.

Let $d = |D_G|$. Each IRI or alternative syntax node creates $O(d^2)$ gates. Each sequence node creates at most one union gate per endpoint pair and one product gate per endpoint triple (u, w, v) , hence $O(d^3)$ gates. The syntax tree has $O(|e|)$ nodes, so C_e has $O(|e|d^3)$ gates. Every edge points to a proper subexpression or a leaf; expression depth is a topological order. $\square$

PROOF OF THEOREM 4.11. By Lemma 4.10, $v_W(\llbracket b_{uv}^e \rrbracket) = \top$ exactly when $(u, v) \in \text{Rel}_e(W)$. For an e -walk $\pi = (v_0, \dots, v_\ell)$ in E_s , let

$$\psi(\pi) = \bigwedge_{j=1}^{\ell} \llbracket b_{v_{j-1}v_j}^e \rrbracket.$$

We prove by induction on $i \geq 1$ that

$$\llbracket r_i(v) \rrbracket = \bigvee_{\substack{\pi: s \rightsquigarrow v \\ |\pi|=i}} \psi(\pi). \quad (14)$$

For $i = 1$, Equation (5) is the disjunction over all one-edge e -walks from s to v . Assume Equation (14) for i . Equation (6) gives

$$\begin{aligned} \llbracket r_{i+1}(v) \rrbracket &= \bigvee_{(u,v) \in E_s} (\llbracket r_i(u) \rrbracket \wedge \llbracket b_{uv}^e \rrbracket) \\ &= \bigvee_{(u,v) \in E_s} \bigvee_{\substack{\rho: s \rightsquigarrow u \\ |\rho|=i}} (\psi(\rho) \wedge \llbracket b_{uv}^e \rrbracket) \\ &= \bigvee_{\substack{\pi: s \rightsquigarrow v \\ |\pi|=i+1}} \psi(\pi). \end{aligned}$$

Appending (u, v) to a walk of length i ending at u gives every walk of length $i + 1$ ending at v , and every such walk arises this way from its own prefix.

Let $n = |V_s|$. If $v \neq s$, a shortest positive e -walk from s to v contains no repeated vertex and has length at most $n - 1$: removing the segment between equal vertices would otherwise yield a shorter walk. If $v = s$, the same deletion argument makes a shortest positive closed walk a simple cycle of length at most n . The construction runs $L \leq n$ levels, and Equation (7) gives

$$\llbracket R(v) \rrbracket = \bigvee_{\substack{\pi: s \rightsquigarrow v \\ 1 \leq |\pi| \leq L}} \psi(\pi).$$

If the construction stopped at an empty level, no walk exceeds L , since a walk of length $L+2$ has a prefix of length $L+1$. Otherwise $L = n$, and removing a repeated segment as above leaves a walk whose edges are a subset of the original, so the longer term implies the shorter one and terms above n add nothing to the disjunction. $R(v)$ therefore contains a satisfied witness for every reachable endpoint. Conversely, every satisfied term denotes an e -walk in W . This proves Equation (8). For e^* , $g_\top$ supplies the zero-length case and $R(v)$ supplies every positive-length case.

Each of the n levels has at most $|V_s|$ union gates and $|E_s|$ product gates, giving $O(n(n + |E_s|))$ gates. One further union gate per endpoint forms $R(v)$. Level $i + 1$ references only C_e and level- i gates, so the level order is topological. Lemma 4.10 gives the global total $O(|e| |D_G|^3 + n(n + |E_s|))$. For $e = p$, the evaluated source-restricted base circuit contains one leaf and at most one union gate for each edge in E_s , so it has $O(|E_s|)$ gates and the total remains $O(n(n + |E_s|))$. $\square$

PROOF OF LEMMA 4.12. Let $C = (\alpha, e^\diamond, ?y)$, let $s = \text{src}_\rho(\alpha)$, and let $\mu = \rho \cup \{?y \mapsto v\}$. If $\diamond = +$, Item 2 of Definition 4.7 assigns $g_C(\rho, v) = R(v)$. For every $W \subseteq G$, Theorem 4.11 and Equation (2) give

$$\begin{aligned} v_W(\llbracket g_C(\rho, v) \rrbracket) = \top &\iff (s, v) \in \text{Rel}_{e^+}(W) \\ &\iff v_W(\text{BProv}_C(\mu)) = \top. \end{aligned}$$

If $\diamond = *$ and $v \neq s$, the same derivation applies with e^* . If $\diamond = *$ and $v = s$, the construction uses $g_\top$, so

$$v_W(\llbracket g_C(\rho, v) \rrbracket) = v_W(\top) = \top = v_W(\text{BProv}_C(\mu))$$

for every W , because the zero-length path exists in every world. Thus, the two Boolean functions agree under every valuation and are equal in $\mathcal{B}(X)$. $\square$

PROOF OF THEOREM 4.13. For a pattern P and mapping μ , let

$$\Phi_P(\mu) = \bigvee_{g \in \text{Cand}_P(\mu)} \llbracket g \rrbracket, \quad \bigvee \emptyset = \perp. \quad (15)$$

We prove $\Phi_P(\mu) = \text{BProv}_P(\mu)$ by structural induction on P .

Triple pattern. For a matching triple t , $\text{Cand}_P(\mu) = \{x_t\}$ and $\Phi_P(\mu) = x_t = \text{BProv}_P(\mu)$. For a nonmatch, both sides are $\perp$.

Closure atom. Let $P = C = (\alpha, e^\diamond, ?y)$ and write $\mu = \rho \cup \{?y \mapsto v\}$ for its input mapping $\rho \in \mathcal{I}_C$ and returned endpoint v . Item 2 of Definition 4.7 produces the corresponding root gate, and Lemma 4.12 gives

$$\Phi_C(\mu) = \llbracket g_C(\rho, v) \rrbracket = \text{BProv}_C(\mu).$$

If no endpoint row is returned, Equation (2) is false under every valuation and both sides are $\perp$.

Filter. Let $P = \text{FILTER}_\varphi(P')$. If $\mu \models \varphi$, then $\text{Cand}_P(\mu) = \text{Cand}_{P'}(\mu)$ and the induction hypothesis gives $\Phi_P(\mu) = \text{BProv}_{P'}(\mu) = \text{BProv}_P(\mu)$. If $\mu \not\models \varphi$, the retained filter yields $\text{Cand}_P(\mu) = \emptyset$ and annotated evaluation multiplies by 0; both sides are $\perp$.

Join. Let $P = \text{JOIN}(P_1, \dots, P_k)$. The candidate product gates range over compatible decompositions $\mu = \mu_1 \cup \dots \cup \mu_k$ and choices $g_i \in \text{Cand}_{P_i}(\mu_i)$. Therefore,

$$\begin{aligned} \Phi_P(\mu) &= \bigvee_{\mu=\cup_i \mu_i} \bigvee_{g_i \in \text{Cand}_{P_i}(\mu_i)} \bigwedge_{i=1}^k \llbracket g_i \rrbracket \\ &= \bigvee_{\mu=\cup_i \mu_i} \bigwedge_{i=1}^k \left(\bigvee_{g_i \in \text{Cand}_{P_i}(\mu_i)} \llbracket g_i \rrbracket \right) \\ &= \bigvee_{\mu=\cup_i \mu_i} \bigwedge_{i=1}^k \text{BProv}_{P_i}(\mu_i) = \text{BProv}_P(\mu), \end{aligned}$$

where the first and last disjunctions range over compatible decompositions, the second equality uses distributivity, and the third uses the induction hypotheses.

Union. Let $P = \text{UNION}(P_1, \dots, P_k)$. All rows for μ bind the same gate g_P , with child set $\bigcup_i \text{Cand}_{P_i}(\mu)$. Thus,

$$\Phi_P(\mu) = \llbracket g_P \rrbracket = \bigvee_i \Phi_{P_i}(\mu) = \bigvee_i \text{BProv}_{P_i}(\mu) = \text{BProv}_P(\mu).$$

If every child collection is empty, no root row exists and both sides are $\perp$.

Projection. Let $P = \text{PROJ}_W(P')$. For an output mapping μ , the child set of g_P is $\bigcup_{\mu': \mu'|_W = \mu} \text{Cand}_{P'}(\mu')$. Hence,

$$\Phi_P(\mu) = \bigvee_{\mu': \mu'|_W = \mu} \Phi_{P'}(\mu') = \bigvee_{\mu': \mu'|_W = \mu} \text{BProv}_{P'}(\mu') = \text{BProv}_P(\mu).$$

Difference. Let $P = \text{DIFF}(P_1, P_2)$ and let g_P, s_P be the gates of Definition 4.7. For every candidate μ of P_1 ,

$$\begin{aligned} \text{pos}(g_P) &= \text{Cand}_{P_1}(\mu), \\ \text{ch}(s_P) &= \bigcup_{v \sim \mu} \text{Cand}_{P_2}(v), \\ \text{neg}(g_P) &= s_P. \end{aligned}$$

Therefore,

$$\begin{aligned} \Phi_P(\mu) &= \left(\bigvee_{g \in \text{Cand}_{P_1}(\mu)} \llbracket g \rrbracket \right) \wedge \neg \left(\bigvee_{v \sim \mu} \bigvee_{h \in \text{Cand}_{P_2}(v)} \llbracket h \rrbracket \right) \\ &= \text{BProv}_{P_1}(\mu) \wedge \neg \bigvee_{v \sim \mu} \text{BProv}_{P_2}(v) = \text{BProv}_P(\mu). \end{aligned}$$

Unfiltered OPT follows from its union-of-join-and-difference expansion.

We prove Equation (11) by induction on the same grammar. Equation (1) proves the triple-pattern case, and Lemma 4.12 with Equation (2) proves the closure-atom case. For a filter, both sides retain μ exactly when $\mu \models \varphi$. For a join, v_W satisfies the circuit formula exactly when one compatible decomposition satisfies every operand formula, which is the support condition for the SPARQL join. Union and projection use disjunction over branches and projected preimages, respectively. Difference satisfies the circuit

formula exactly when the left mapping occurs and no compatible right mapping occurs. These cases exhaust the supported grammar.

The induction proves Equation (10) for the outer projection root. If no candidate row exists, Equation (15) gives $\perp$. Proper-subpattern dependencies and the path-level order yield a topological order. Proposition 4.9 proves the statement for factored basic-pattern subplans. $\square$

PROOF OF COROLLARY 4.14. Let $g = \text{root}(\mu)$. Equation (11) gives

$$v_W(\llbracket g \rrbracket) = \top \iff \mu \in \text{supp}(\llbracket Q \rrbracket_W)$$

for every possible world W . Therefore,

$$\Pr(\mu) = \sum_{W \subseteq G} 1[v_W(\llbracket g \rrbracket) = \top] \prod_{t \in W} \Pr(x_t) \prod_{t \notin W} (1 - \Pr(x_t)). \quad (16)$$

The right-hand side is the weighted model count of $\llbracket g \rrbracket$. d-DNNF compilation preserves the Boolean function, and WMC returns Equation (16) in time linear in the compiled size [1]. $\square$

REFERENCES

- [1] Adnan Darwiche and Pierre Marquis. 2002. A Knowledge Compilation Map. *Journal of Artificial Intelligence Research* 17 (2002), 229–264. doi:10.1613/jair.989